

A-Grid:

SmartGrid Monitoring & Fault Detection System

using AI and PoA² Blockchain

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Executive Summary

A-Grid is a hybrid AI-blockchain framework that delivers real-time intelligence and data integrity to modern power grid infrastructure. As global energy networks become more complex, the systems that monitor them haven't kept up, leaving utilities vulnerable to billions in losses from undetected faults, energy theft and tampered meter data.

A-Grid fills these gaps with three tightly coupled layers: Edge Intelligence Layer, powered by machine learning, for millisecond-level fault detection. Data Trust Layer, to validate sensor data before it ever reaches the blockchain. Blockchain Trust Layer, using PoA² consensus to create an immutable, auditable record of all system events.

This proposal outlines the technical architecture, the implementation roadmap, the performance targets and the commercial potential of the A-Grid system designed for rapid prototyping and real-world validation.

1. Background and Problem Statement

Modern power systems face critical challenges in energy monitoring, fault detection, and data trust losses over \$96B annually to Non-Technical Loss (NTL) and undetected faults. Current monitoring solutions suffer from:

Data Integrity Problem (Oracle Issue): 90% of Smart meters data can be manipulated before reaching the system.

Response Latency: Centralized systems take minutes to identify faults that cause damage in milliseconds.

Lack of Trusted Records: No tamper-proof system for auditing faults and energy data

Scalability Issues: Traditional systems struggle with real-time data processing

2. Proposed Solution: A-Grid Framework

A-Grid introduces a hybrid AI-blockchain smart grid monitoring system that ensures:

2.1. Data Trust Layer

- Software-based validation + optional hardware trust (TEE-ready design)
- Data is verified before blockchain submission
- Future-ready for hardware security integration

2.2. Edge Intelligence Layer (Core Innovation)

- AI-based anomaly detection (Isolation Forest / LSTM)
- Real-time fault detection at the edge (milliseconds level)
- Detects:
 - Voltage anomalies
 - Energy theft patterns
 - Equipment faults

3. Blockchain Trust Layer

- Uses PoA² consensus via Pure chain Open Cloud
- Records:
 - Verified data (hash)
 - Fault events
 - System state
- Ensures:
 - Immutability
 - Traceability
 - Low latency and high TPS

4. System Architecture

4.1. IoT / Data Layer

- Smart meters (or simulated devices for prototype)
- Measures:
 - Energy production
 - Energy consumption
 - Voltage / current

4.2. Edge & Middleware Layer

- Built using Node.js / Python
- Performs:
 - Data validation
 - AI-based fault detection
 - Data filtering

4.3 Storage Layer

- IPFS (off-chain):
 - Stores full sensor data
- Blockchain (on-chain):
 - Stores hash + metadata

4.4 Blockchain Layer

- Purechain PoA² (via API)
- Handles:
 - Transaction validation
 - Consensus
 - Immutable recording

4.5 Application Layer

- Dashboard (optional for prototype)
- Displays:
 - Energy usage
 - Fault alerts
 - Historical data

5. Technical Metrics & Performance Analysis

Metric	Target
Fault Detection Latency	< 100 ms (Edge AI)
Blockchain Finality	< 2–3 seconds
Throughput	500–2000 TPS (PoA ²)
Storage Reduction	> 70% (IPFS hybrid)

6. AI-Based Fault Detection (Practical Version)

Phase 1 (Prototype):

- Isolation Forest (fast, lightweight)
- Phase 2 (Research Extension):
- LSTM for time-series

prediction Capabilities:

- Detect anomalies in real-time
- Identify abnormal consumption patterns
- Improve detection accuracy over time

7. Implementation Roadmap

The project is structured across four phases, each with clear deliverables and success criteria:

Phase	Title	Key Deliverables
Phase 1	Prototype Development	Set up backend (Node.js + Python AI), integrate PoA ² API, simulate smart meter data
Phase 2	System Integration	Connect AI + backend + blockchain, integrate IPFS storage, implement validation layer

Phase 3	Testing & Evaluation	Measure latency, TPS, and detection accuracy across all components
Phase 4	Optimization & Demo	Improve performance, add monitoring dashboard, prepare live demonstration

8. Innovation and Contribution

A-Grid introduces:

- AI-based real-time fault detection
- Multi-layer validation (before blockchain)
- Hybrid storage (IPFS + blockchain)
- Efficient consensus using PoA²
- Cyber-physical smart grid monitoring

9. Commercial Viability & Impact (The ROI)

- Utility Benefit: Expected 15–20% reduction in Non-Technical Loss within the first 12 months.
- Insurance Value: "Forensic-as-a-Service" model where insurers pay for access to the immutable fault ledger to verify claims.
- Incentive Model: A P2P "Energy Trust" score. Prosumers with high scores (verified green production) receive lower transaction fees on the network.
- Support future smart grid infrastructure.

References

- IEEE 2030.5: The standard for Smart Energy Profile (SEP) communications.
- NIST IR 7628: Guidelines for Smart Grid Cybersecurity.
- IEC 61850: The international standard for communication protocols in substations.
- ISO/IEC 27001: For the information security management system of your blockchain nodes.

